

Work, Force and Kinetic Energy

SimuPhysics Work-Energy Theorem · SimuPhysics

Years 8-9

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://simuphysics.com/view/mechanics-work-energy-theorem.html>

Curriculum links: AC9S8U05 · AC9S9U05 · energy transfer

Learning intentions

- I can calculate the work done by a force over a distance.
- I can compare calculated work done with the change in kinetic energy shown in the simulation.
- I can explain the work-energy theorem in my own words using evidence from the simulation.

Getting started

Open the SimuPhysics Work-Energy Theorem simulation. Set an object in motion by applying a constant force over a chosen distance, noting the mass, force, distance and starting speed values before you press go.

Tasks

1. Apply a force of 10 N over a distance of 5 m to an object starting at rest. Record the object's final speed and kinetic energy shown by the simulation.

2. Calculate the work done by this force using $W = F \times d$. Compare your answer to the change in kinetic energy shown in the simulation.

Working:

3. Keeping the force the same, increase the distance (e.g. to 10 m). Record the new final kinetic energy, then calculate the work done and check it matches.

Working:

4. Try three different force values (e.g. 5 N, 10 N, 20 N) over the same fixed distance. Record the work done and final kinetic energy for each.

Force (N)	Work done (J)	Final KE (J)

5. Using your table, describe the relationship between the size of the force applied and the kinetic energy gained by the object.

6. Explain the work-energy theorem in your own words: what does the work done on an object tell you about its change in kinetic energy?

7. The simulation assumes no friction or air resistance. Explain how the results might be different for a real object, like a shopping trolley, being pushed the same distance with the same force.

Extension (optional)
A car applies twice as much braking force as another identical car, over the same stopping distance. Use the work-energy theorem to explain what this tells you about the difference in their initial kinetic energies (and therefore their initial speeds).